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قسم هندسة نظم أمن المعلومات الذكية

تحسين إمكانية كشف التطفل ضمن المنازل الذكية

Improving Intrusion Detection in Smart Homes

تقرير مشروع تخرج 1 قِدّم استكمالاً لمتطلبات الحصول على درجة الإجازة في هندسة الذكاء الاصطناعي – هندسة نظم أمن المعلومات الذكية

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Abstract

This project presents an Intrusion Detection System (IDS) for smart home IoT environments using supervised machine learning. The system is designed to identify devices, detect malicious activities, and classify different types of attacks.

A multi-stage architecture with three models was used: device identification, binary attack detection, and multi-class attack classification. The models were trained using TON_IoT and CICIoT datasets.

Data preprocessing and balancing techniques were applied to improve performance. Random Forest and XGBoost were used for classification tasks

The results show high performance, with some limitations in detecting certain attack types such as spoofing.

الملخص

مع التطور السريع لإنترنت الأشياء (**IoT**)، أصبحت بيئات المنازل الذكية أكثر انتشارًا، حيث تتكون من مجموعة من الأجهزة المترابطة مثل الحساسات والأجهزة الذكية التي تتبادل البيانات بشكل مستمر عبر الشبكة.

ومع ذلك، تعاني أجهزة **IoT** من محدودية الموارد وضعف في آليات الحماية، مما يجعلها عرضة للعديد من الهجمات الإلكترونية مثل هجمات حجب الخدمة (**DoS**)، والانتحال (**Spoofing**)، وهجمات الرجل في الوسط (**MITM**). كما أن أنظمة كشف التطفل التقليدية لا تكون مناسبة دائمًا لهذه البيئات بسبب طبيعتها الديناميكية وتنوع الأجهزة.

لذلك، يقترح هذا المشروع نظام كشف تطفل يعتمد على تعلم الآلة، مخصص لبيئة المنازل الذكية. يقوم النظام بتحليل سلوك الأجهزة وحركة الشبكة بهدف اكتشاف الأنشطة الخبيثة.

يعتمد النظام على بنية متعددة المراحل، حيث يتم أولاً تحديد نوع الجهاز، ثم الكشف عن وجود هجوم، وأخيرًا تصنيف نوع الهجوم، مما يساعد في تحسين دقة الكشف وفهم التهديدات بشكل أفضل

Index Terms—Internet of Things (IoT), Smart Homes, Networking, Security, Intrusion Detection, Anomaly Detection, Supervised Machine Learning, Classification, Heterogeneity

Introduction:

The popularity of Internet of Things (IoT) devices has significantly increased over the past few years. This is due to their ubiquitous connectivity, allowing them to communicate and exchange information with other technologies, their intelligence, and their decision-making capabilities. These features provide seamless user experiences and significantly enhance people's everyday lives. The rapid growth of IoT is not limited to smart homes, but also extends to critical domains such as smart cities, healthcare systems, intelligent transportation, and industrial infrastructures.

However, despite their advantages, IoT devices introduce serious security challenges. Their dependence on Information and Communication Technology (ICT), combined with weak security mechanisms, makes them highly vulnerable to various cyberattacks such as Denial of Service (DoS), spoofing, and man-in-the-middle (MITM) attacks. These attacks can cause severe consequences, including service disruption, data leakage, system failure, and even physical harm in critical applications.

One of the main reasons for these vulnerabilities is the limited computational resources of IoT devices, including constrained processing power, memory, and energy. Additionally, the heterogeneity of IoT environments, in terms of hardware, software, and communication protocols, makes it difficult to design and deploy effective and scalable security solutions. Traditional security mechanisms, such as firewalls and signature-based intrusion detection systems, are not well suited for IoT

environments, as they are typically static and cannot adapt to the dynamic and diverse nature of IoT networks.

Moreover, IoT devices often operate deep inside the network, which makes traditional perimeter-based defenses insufficient. As the number of connected devices continues to grow, the number of potential vulnerabilities and attack surfaces also increases, further weakening traditional security approaches. Therefore, there is a clear need for intelligent and adaptive security systems capable of detecting malicious activities in IoT environments.

In recent years, several studies have focused on developing intrusion detection systems (IDS) for IoT networks. However, most existing approaches are limited to detecting specific types of attacks, such as DoS, and do not provide comprehensive solutions. In addition, many systems fail to consider device behavior profiling or to accurately classify the type of attack. This limits their effectiveness in real-world scenarios, where attacks are more complex and diverse.

To address these challenges, this project aims to develop an Intrusion Detection System (IDS) for smart home IoT environments using supervised machine learning techniques. The proposed system is designed to identify IoT devices, detect malicious activities, and classify different types of cyberattacks.

Chapter One: Theoretical Framework

Section One: Internet of Things (IoT)

The Internet of Things (IoT) is creating a new world — a measurable and monitorable world — in which people can manage their lives more effectively, cities can manage their infrastructure more efficiently, and companies can operate their

businesses more productively. This connected smart world will introduce fundamental changes for consumers and society, while also transforming entire industries. The rise of IoT will bring many important improvements to our world and daily lives by helping us make better and faster decisions based on real-time, high-quality information.

What is the Internet of Things?

To understand the Internet of Things, it is useful to briefly review how the Internet itself evolved.

The Internet initially started as a relatively small network of military and governmental computers connected through wired communication links. Over time, the Internet expanded and became accessible to millions of people around the world. The World Wide Web made it possible for anyone to publish information for public access.

The next major development came with the rapid spread of wireless communications. With the help of Wi-Fi and cellular networks, Internet access expanded dramatically. Smartphones and portable Internet-connected devices freed Internet access from fixed locations, making connectivity available almost everywhere.

However, computers were not the only devices using the Internet. Many other types of devices gradually began to include simple computing and networking capabilities. In the early 2000s, the term Internet of Things (IoT) emerged to describe the continuously growing range of Internet-connected objects and their applications. In 2005, the International Telecommunication Union (ITU) officially recognized the term and defined IoT as:

"A global infrastructure consisting of smart objects connected to the Internet that communicate with each other with minimal or no human intervention."

These objects usually connect through a Gateway to an IoT platform, which consists of software tools and services that collect data from sensors, controllers, and other devices.

Today, IoT focuses on connecting many different types of devices to the Internet, such as sensors, hands-free controllers, and remote monitoring systems. According to Cisco, the number of Internet-connected "things" has already exceeded the number of people connected to the Internet, and in fact, this milestone was reached more than a decade ago.

Figure 2-1 illustrates the general IoT architecture. In the Sense & Connect Layer, the platform links IoT hardware with network applications responsible for data processing and storage. In the middle lies the Platform Layer, where data is stored, secured, processed, and analyzed. Finally, the End-User Layer connects to user-oriented applications that monitor and interpret data received from IoT devices, then send commands instructing devices to perform actions such as locking a house door or opening a garage door.

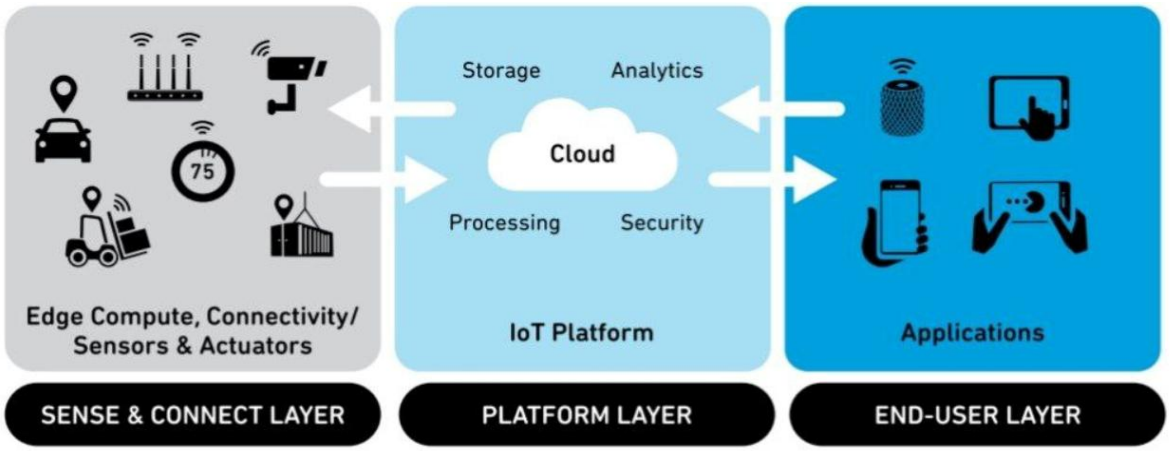

Figure 0-1Internet of Things Layers

IoT platforms usually provide services characterized by the following features:

Adaptability: IoT systems can dynamically adapt to changing contexts and behave according to operating conditions, user inputs, or sensed information.

Unique Identification: Each IoT device has a unique identity, such as an IP address. These addresses allow users to query, monitor, and remotely control devices.

Self-Configuration: Devices within an IoT network can automatically configure themselves, enabling multiple connected devices to cooperate in providing complete system functionality.

Network Integration: IoT devices are integrated into the IoT network to enable communication between nodes, gateways, and infrastructure components.

Intelligent Decision Making: IoT devices are capable of making decisions that allow them to adapt to changing conditions.

Wireless Standards Enabling the Internet of Things

A wide range of wireless communication standards are currently used to connect IoT devices. Some of these standards are well known, such as Bluetooth, Wi-Fi, and Zigbee, while others rely on less common proprietary solutions.

Figure 1-3 summarizes the current landscape by dividing it into three sections based on communication range. As illustrated in the figure, the IoT ecosystem has become complex and difficult to manage from the user's perspective due to the large number of different standards being used.

Companies developing IoT products usually select the communication standard (or combination of standards) based on factors such as communication range and data type. For example, devices may need to transmit media content or sense-and-

control data. With a large variety of standardized and proprietary technologies available, system designers can choose the most suitable communication technology to optimize end-user device performance. However, this does not always make things easier for ordinary users.

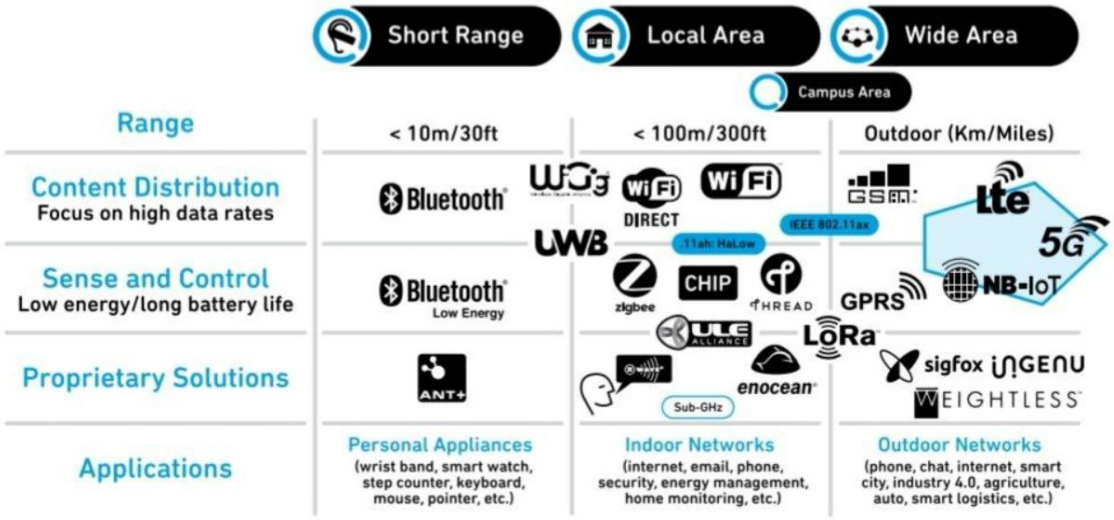

Figure 1-0-2 Early Stages of the Internet of Things

Goal: **Interoperability**

Interoperability refers to the ability of a product or system to work with other products or systems, both now and in the future, without restrictions.

Consumers want all of their wireless home devices to communicate with each other in a "plug-and-play" manner, without worrying about whether a new IoT device they purchase will work with their existing network. They expect to buy any wireless product and have it function automatically. However, this is not always possible today.

For example, purchasing a smart door lock or an Internet-connected light bulb can become complicated because users must consider how the device will communicate

with other smart home devices, including those using Alexa, Google Assistant, or Siri.

Some manufacturers rely on standard wireless technologies to provide connectivity, but they also add proprietary solutions to control and interact with IoT devices. However, this approach creates challenges for users, especially during the purchasing or installation process, because new devices may not integrate smoothly with existing networks as expected.

Examining IoT Architecture

Some IoT applications are already a reality today, such as controlling a home thermostat wirelessly or using a smartphone to unlock car doors. However, future IoT applications are expected to be much broader and more advanced.

These future applications will require IoT platforms capable of performing massive data transmission and background processing operations.

An IoT platform/architecture consists of several interconnected layers (see Figure 1-1). The following provides a closer look at the individual components within these layers:

Sensors and Actuators:

IoT sensors and actuators measure factors such as temperature, sound, humidity, and vibration.

In a typical smart home, a smart device such as a thermostat includes a built-in communication module connected to the home network. These sensors and actuators convert physical measurements into electrical values that operate the system.

IoT Gateways:

The gateway transfers data between the local network and the Internet. Electrical values collected from sensors and actuators are uploaded to the local network using networking protocols such as Bluetooth, Bluetooth Low Energy (BLE), Cellular, LoRaWAN, Thread, Wi-Fi, or Zigbee.

The gateway creates a meshed network backbone to distribute collected data and send responses back to devices.

Cloud-based IoT Platform

The data transmitted through the gateway is stored and processed within a cloud-based IoT platform or company data center. This data is then used to perform intelligent actions and support decision-making.

Applications:

Finally, data generated by IoT devices is used within applications to help individuals or organizations make better decisions or perform specific actions. These applications deliver information from the cloud to smartphones, tablets, or computers.

The application layer is considered the most important layer for users because it represents their interface with the IoT network, enabling them to monitor and control various system components, sometimes in real time.

IoT Protocols:

Communication protocols form the backbone of IoT systems, connecting IoT devices to the network and ultimately to applications and users. These protocols, whether standardized or proprietary, enable data exchange between the different

layers of IoT architecture by defining data formats, encoding methods, device addressing mechanisms, and the way packets are routed from nodes to destinations.

The IoT ecosystem includes a wide variety of protocols that support short-range, local-area, and wide-area networks, all coexisting together. Each technology has specific characteristics in terms of range, sensing and control capabilities, and support for different types of information transmission (Figure 1-2).

Together, these technologies provide seamless coverage across different wireless ranges and capabilities. For example, Bluetooth works efficiently in short-range applications, while Narrowband IoT (NB-IoT) is suitable for long-range applications.

Summary of Major IoT Networking Protocols:

IEEE 802.15.4: is a set of standards for Low-Rate Wireless Personal Area Networks (LR-WPAN). These standards provide low-cost and low-speed communications for low-power devices. They form the basis for higher-level communication protocols such as Zigbee.

Zigbee is a low-power mesh network designed for smart homes and smart energy applications. It is based on the PHY and MAC layers of IEEE 802.15.4.

Wi-Fi: is a family of wireless local area networking (WLAN) standards based on IEEE 802.11. It provides high data rates for both indoor and outdoor environments and is widely used worldwide.

Bluetooth

Bluetooth is an open standard managed by the Bluetooth Special Interest Group (SIG). It is a low-cost wireless communication technology suitable for short-range data transmission between mobile devices, typically within 8 to 10 meters.

Bluetooth is commonly used in audio streaming, vehicles, speakers, and headsets.

Bluetooth Low Energy (BLE)

BLE is part of the Bluetooth standard specifically designed to minimize power consumption. BLE devices often use small coin-cell batteries.

BLE is commonly used in IoT devices such as smart light bulbs and smart switches.

Thread: is a secure, low-power mesh networking technology based on Internet protocols and designed specifically for IoT products.

In 2014, the Thread Group was established to support the adoption of this technology. Thread supports IPv6-based communication standards within its secure and low-power network.

LoRa: is a Low-Power Wide-Area Network (LPWAN) protocol developed by the LoRa Alliance. It is ideal for enabling sensors in large-scale agricultural applications because it supports long-range communication with low power consumption.

Cellular Standards: Cellular technologies such as 5G provide the backbone for IoT networks, supporting both high data rates and long-range communication.

One of the most widely used standards in this field is NB-IoT, which is used in smart parking systems, utility management, and manufacturing automation.

These messaging protocols are used to share data between devices and the cloud. IoT protocols are a fundamental part of the IoT technology stack; without them, physical devices would not function properly.

IoT protocols enable devices to exchange data in a controlled and meaningful manner.

For example, IoT devices are considered “smart” because they not only communicate with each other, but can also automatically mitigate problems or request assistance through the network when issues occur. Such interaction is only possible through communication protocols, which serve as the common language between IoT devices.[1]

Section Two: Smart Homes

In this section, we explore how the Internet of Things supports smart homes and how mesh/unified networking technologies enable devices in every room to connect to IoT systems. This section also discusses some of the challenges facing IoT and how emerging technologies help address them.

The State of Smart Homes Today

The concept of the smart home has gained significant popularity in recent years, as manufacturers have developed a growing range of devices capable of connecting to home networks.

Smart home networks enable homeowners to use resources more efficiently, control devices and appliances through voice commands, and make decisions based on the data received from these devices. Smart homes offer enormous potential, as they can improve energy efficiency, comfort, security, entertainment, and overall quality of life.

Most smart homes today are equipped with multiple IoT devices, hubs, and sensors that communicate through the home network or gateway and ultimately connect to the cloud. Homeowners can control these systems from anywhere inside the house or remotely through the Internet.

Smart home systems may include several wireless communication standards related to home applications. The table below illustrates the characteristics and differences between some of these standards. Furthermore, these home networks may consist of multiple interconnected subnetworks operating under a main gateway.

Wireless Protocol	Wi-Fi	Zigbee/Thread	Bluetooth
Standard	IEEE 802.11	IEEE 802.15.4	IEEE 802.15.1
Frequency band	2.4, 5, 6 GHz	2.4 GHz	2.4 GHz
Nominal range	150 meters	100 meters	10 meters
Data rate	11 Gbps	250 Kbps	1 Mbps

Figure 1-0-3 Smart Home Standards

Many smart homes today consist of a dedicated set of nodes. A node is a piece of hardware that has a unique address on the network; a node can either be a device or a hub:

Devices: A device (such as a thermostat or a light bulb) usually has only one connection to the network. Depending on its function, the device may or may not need to track real-time information.

Hubs: A hub is a device that manages network traffic. It may have multiple connections because it communicates with several devices. Hubs must track real-time information in order to efficiently and effectively transmit events.

A single-hub network consists of one hub node and one or more device nodes. The hub manages the exchange of events and information between devices on the network.

A multi-hub network contains more than one type of hub, as shown in the following figure. The network diagram illustrated in the figure contains two hubs, each using a different networking protocol (Bluetooth versus Zigbee). Individual devices connect to one of the hubs, and both hubs then connect to the gateway, which is capable of communicating with both of them.

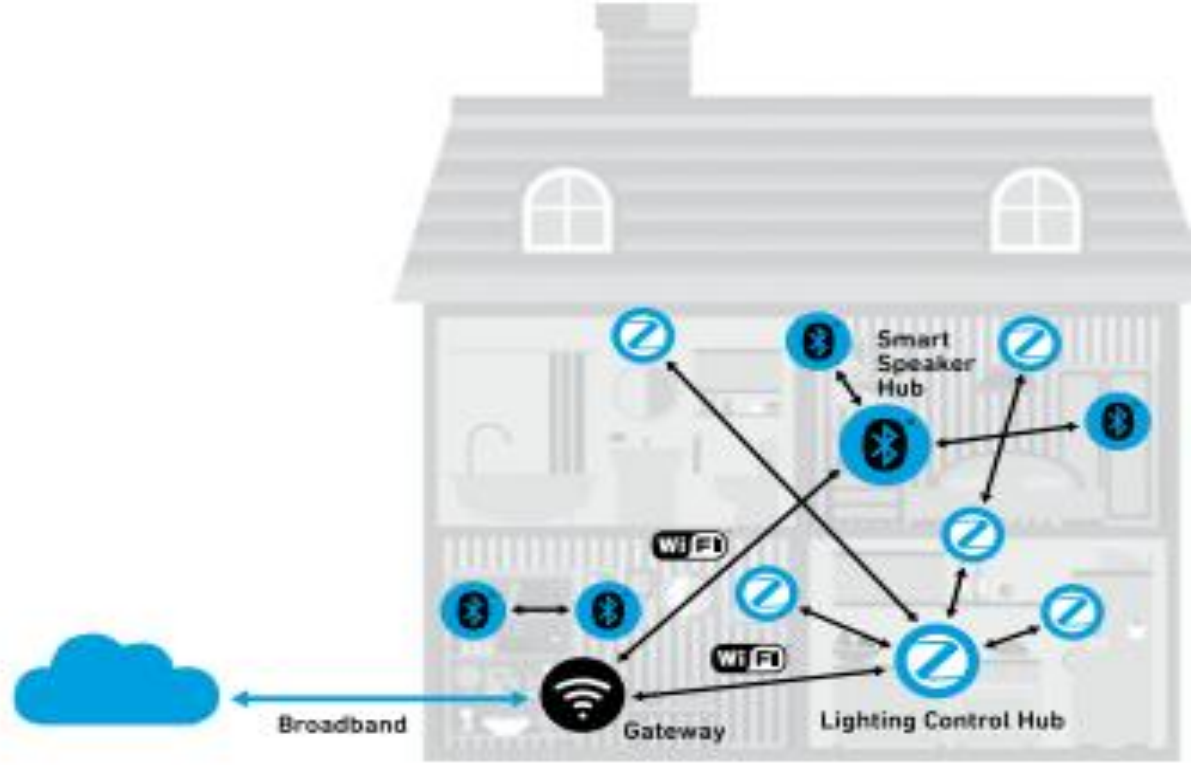

Figure 1-0-4 Smart Home with Zigbee Hub and Bluetooth Hub

Mesh Networking / One Pod per Room

To create a smarter and more connected home with a more reliable communication network, consumers are moving toward a newer topology known as mesh networking, sometimes referred to as "one pod per room."

Mesh networks easily extend the range of wireless devices and provide high and consistent data rates throughout homes, offices, and even outdoor environments. In addition, mesh networks that support "one pod per room" with multi-protocol capabilities can reduce the cost and complexity of having multiple separate networking systems within the home.

Mesh networking represents an important advancement in the evolution of Wi-Fi technology, especially for smart homes. Mesh Wi-Fi systems typically include a router connected to the main Internet connection, along with satellite router-like devices or nodes distributed throughout the home to provide complete Wi-Fi coverage for smart home IoT devices (see figure).

These advanced mesh networks provide full redundancy and maximize overall network performance.

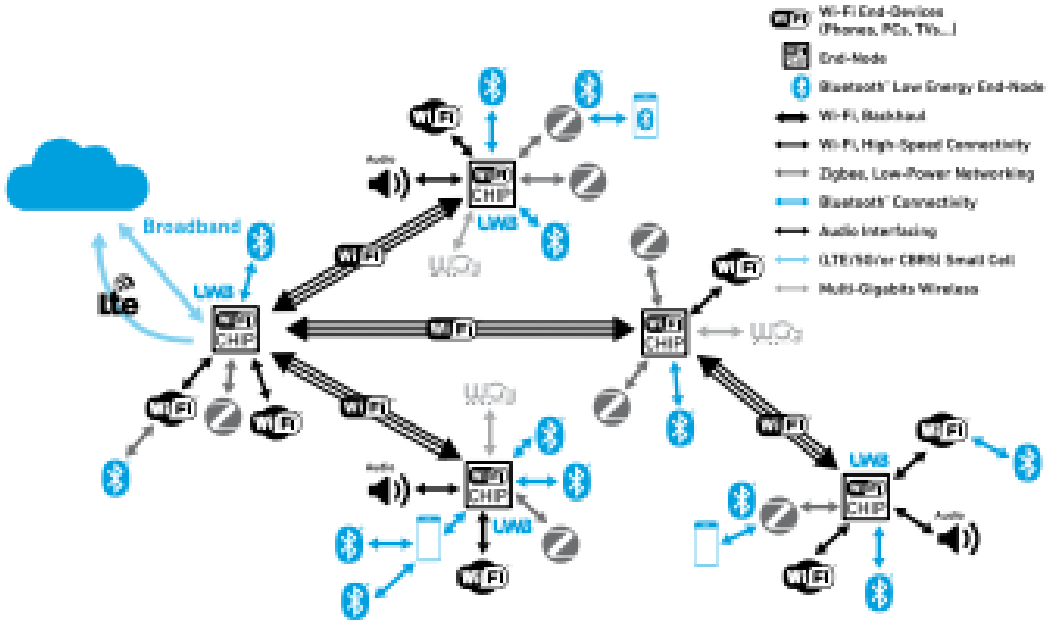

Figure 1-0-5 IoT Communication Network Distributed Using One Pod per Room

A mesh network can automatically configure itself. Therefore, it can integrate a new node into the existing network without requiring manual modifications from the network administrator. This makes the network more adaptable and scalable. In addition, a mesh network can self-heal by finding the fastest and most reliable signal paths for data transmission, even if one of the nodes is blocked or lost.

Many modern mesh networks use what are known as tri-band Wi-Fi gateways and edge nodes. Tri-band Wi-Fi improves data capacity and coverage by using the 2.4 GHz, 5.2 GHz, and 5.6 GHz frequency bands within gateways and edge nodes. In a tri-band system, the second 5.6 GHz band acts as a dedicated communication link between routers on the network, increasing overall system speed by up to 180% compared to traditional dual-band systems.

The advantages of tri-band Wi-Fi are therefore clear. First, it allows more wireless devices to connect using the faster 5 GHz band and, with Wi-Fi 6E, both the 5 GHz and 6 GHz bands. If the network is a mesh system with multiple routers distributed throughout the environment, the higher-frequency band functions as a dedicated backbone connection between routers to accelerate the entire system.

The third radio in a tri-band system is also used to carry communication between mesh network routers. As a result, it creates a dedicated wireless channel specifically for data transfer between routers. Although dual-band systems use the same Wi-Fi band for both client devices and inter-router communication, tri-band systems dedicate one band for backhaul communication, improving efficiency by up to 50%, especially as more devices are added to the network.

Furthermore, because this radio connection is faster, high-speed Internet services provided by DSL or cable modems are distributed efficiently across the entire

network. This is especially useful for streaming Wi-Fi videos to smart TVs and cameras deployed throughout modern smart homes.

To future-proof a home network, adopting a tri-band system is highly recommended. Gigabit Internet services are becoming increasingly common, and tri-band systems provide the best long-term solution for ensuring future network speed and capacity.

Another major advancement in mesh networking systems is the introduction of filtering technologies to increase the capacity and range of the "one pod per room" topology. The use of band Boost filters maximizes capacity and coverage by improving band isolation in tri-band systems. Similarly, edge Boost filters enhance capacity and range by maximizing output power near the edges of Wi-Fi frequency bands.

These topics, along with how they address coexistence challenges in mesh networks, will be discussed in greater detail in the following section.

As discussed in Chapter One, major manufacturers are working on a new standard called Connected Home over IP (CHIP) to improve device interoperability and reduce complexity within mesh networks.

During the transition toward CHIP, multi-standard radios will be required. These radios will future-proof the network by supporting migration to CHIP while maintaining backward compatibility with older standards. Although CHIP was still under development at the time and expected to launch around 2021, cooperation between standards organizations and leading manufacturers has accelerated industry adoption.

Overcoming Challenges:

We now discuss some of the major challenges facing smart home IoT network and device manufacturers.

Interoperability: is the ability of systems, applications, and services to work together reliably and predictably. It helps ensure ease of use and accelerates technology adoption.

Users expect their wired and wireless electronic devices to connect easily and operate together with minimal effort, commonly referred to as "plug-and-play." The same expectation applies to smart home networks, which consist of many devices developed by different manufacturers.

Interoperability and certification programs help ensure compatibility between devices using major IoT standards. The widespread success of Wi-Fi is largely due to the interoperability program managed by the Wi-Fi Alliance, which operates one of the world's most trusted certification systems. Tens of thousands of certified Wi-Fi devices provide seamless and proven connectivity.

Today, Zigbee 3.0 offers a similar level of interoperability through partnerships and certification programs. Interoperability between Zigbee and Thread devices is also supported through cooperation between the two groups. In addition, Bluetooth provides certification programs covering both protocol stacks and application profiles, helping Bluetooth and Bluetooth Low Energy (BLE) deliver reliable connectivity experiences.

These alliances and certification programs have played a crucial role in integrating the fragmented, multi-standard smart home IoT ecosystem. Manufacturers developing devices based on these protocols must comply with certification and

alliance requirements before their products can enter the market. This ensures the seamless interoperability users expect.

Radio Frequency Interference :

Radio Frequency (RF) interference in smart home networks has become increasingly complex with each new generation of wireless technologies.

Many wireless home communication devices operate on overlapping or closely adjacent frequency bands. In some cases, these frequency ranges are so close that they cause interference affecting transmission and reception quality.

IoT and 5G technologies add another layer of complexity, increasing the challenges faced by smart home device designers. Modern smart home IoT devices must support multiple RF paths, sometimes across wider frequency ranges, while maintaining low power consumption and preventing interference between standards and transmitted or received data.

With so many standards operating within a single network, reducing interference has become essential.

Maintaining Seamless, **Energy-Efficient**, **and Reliable Connectivity** :

Some companies develop products that support multiple standards, improving interoperability within IoT networks by enabling communication with a broader range of devices.

With dynamic multi-protocol support, a node can communicate with devices using different standards.

For example, a smart home network may include Zigbee and BLE devices. In this case, a mesh node supporting both Zigbee and BLE can automatically switch

communication between these protocols and interact with each standard individually at a given moment.

Qorvo IoT communication products used in smart home devices provide additional capabilities through support for Concurrent Connect technology, which continuously receives and translates communications, as illustrated in Figure 3-6.

This innovation enables faster and more reliable protocol switching, reducing packet loss while maintaining multi-protocol support.

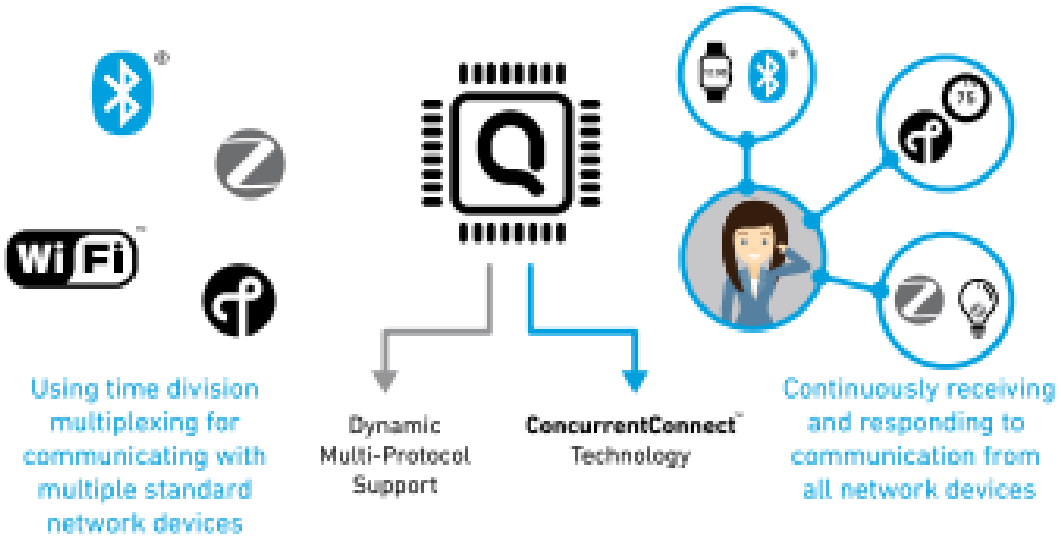

Figure 1-0-6 Dynamic Multi-Protocol Support and Current Connect Technology

For example, as illustrated in Figure 3-6, the left side shows that IoT communication products support dynamic multi-protocol support. This capability allows communication between multiple IoT devices using different standards, enabling switching from one protocol to another whenever needed.

However, Qorvo adds another layer of sophistication to its IoT RF devices through a technology known as Concurrent Connect. This technology makes IoT

communications more reliable and seamless because it enables devices to continuously and simultaneously receive and transmit communications across the network. As a result, switching between protocols becomes seamless, eliminating the need for repeated "ping" operations and preventing potential communication interruptions.

For example, without Concurrent Connect, Zigbee/CHIP and Bluetooth Low Energy (BLE) devices would alternate communication between the two standards, communicating with only one protocol at a time. This protocol switching introduces delays that can interrupt communications. In contrast, devices equipped with Concurrent Connect can instantly switch between BLE and Zigbee/CHIP with little or no interruption at all. This is because Concurrent Connect eliminates protocol-switching delays, making communication faster, more efficient, more scalable, and capable of handling more packets within a shorter period of time.

Building a User-Friendly Self-Managing IoT Network :

Most IoT users expect a "plug-and-play" experience that requires little or no setup. Users simply want devices to work immediately after being powered on. This places the responsibility for usability on manufacturers, which is not always easy because every user's network environment is different.

To achieve this goal, the home IoT mesh network must be self-managing and self-healing so that users can trust it to continue operating without requiring technical intervention. The IoT network should be capable of monitoring itself and notifying users of potential problems before complete network failures occur.

Individual nodes and gateways must cooperate and adapt to network failures or environmental changes. Smart home subsystems continuously monitor signal

strength, battery life, and other parameters, dynamically adapting and notifying users to ensure optimal performance.

Mesh networking provides significant advantages for IoT applications because it is always available and accessible. Mesh networks use self-healing algorithms that automatically select the best route for transmitting data even if some nodes unexpectedly lose connectivity. These algorithms ensure that data is routed only through available and functioning links. Consequently, the mesh network continues operating even when some nodes become inactive.

Additionally, mesh networks allow devices to communicate directly with one another without routing all traffic through the gateway.

Network Security and Privacy :

Implementing adequate security and privacy measures remains a major challenge when deploying IoT systems. Modern IoT networks are often fragmented and rely on a wide variety of protocols and technologies, making security and privacy management difficult.

This situation highlights the importance of standardization, which can improve security, privacy, usability, and scalability while enabling users to add devices to the network more easily and securely.

Using mesh networking, as previously discussed, places greater control and security in the hands of the owner. These mesh systems are usually configured, managed, and monitored through dedicated applications. Such applications help users configure the network, assign secure passwords, manage network issues, check data speeds, and apply parental controls. Some systems also allow owners to monitor who is connected to the network at any time.

Just as software updates are essential for maintaining the security of computers and smartphones, they are equally important for mesh networks. Fortunately, mesh networks often update themselves automatically, relieving users from manually performing disruptive updates.

Some mesh systems also offer paid subscription services that provide enhanced security features, including protection against hackers, malicious websites, and security threats. These subscriptions may also include protection against malware, viruses, hackers, and cybercriminals.

For smart homes equipped with network-connected security cameras, thermostats, and other IoT devices, mesh networking systems provide valuable protection for the entire network against cyberattacks.

Power Consumption:

Fossil fuel resources around the world are decreasing rapidly, increasing pressure to reduce energy consumption and develop more environmentally friendly products. This pressure also applies to manufacturers and suppliers producing IoT devices today, and it is expected to intensify further as global green initiatives expand.

Some IoT products use coin-cell batteries that may last up to ten years, although producing these batteries still consumes fossil fuels. Many IoT devices do not use coin-cell batteries at all but instead draw power directly from the electrical grid. To reduce energy consumption, software and hardware solutions are sometimes implemented to keep devices in sleep mode whenever active communication is unnecessary.

In industrial environments, IoT devices not only improve efficiency and automation but are also expected to consume less energy during operation. Manufacturers are

increasingly focusing on producing devices that are more energy-efficient, smaller, and more elegant in design. This benefits consumers, manufacturers, and companies alike.

IoT technologies can also support renewable energy initiatives. Installing IoT sensors on generation, transmission, and distribution equipment enables remote asset monitoring and management. In manufacturing environments, IoT deployment can reduce costs associated with lighting, heating, cooling, and safety systems.

Smart home networks may contain numerous communication devices operating continuously 24/7. Since these devices consume energy even when not actively transmitting data, efficiency and power consumption remain major design concerns. IoT devices must therefore be optimized for energy efficiency regardless of the protocol they use.

At a high level:

Wi-Fi and CHIP (Zigbee, Thread) provide similar communication range, but Wi-Fi consumes significantly more power because it supports higher data rates.

CHIP (Zigbee, Thread) and BLE consume similar levels of power and were specifically designed for battery-powered operation.

CHIP technologies consume approximately 1% to 3% of the power required by Wi-Fi, depending on the application.

With global green initiatives becoming increasingly important, reducing energy consumption across IoT networks has become essential. Both users and manufacturers seek lower power consumption without compromising performance or connectivity goals. Devices can also improve reliability by automatically notifying users when batteries require replacement.

Importance of Always-On Connectivity:

IoT networks in homes and workplaces have expanded dramatically over the past few years, while our dependence on these networks has also increased. More people now work from home offices, while families rely on home networks for video calls, gaming, streaming entertainment, health monitoring, and home security.

Reliable, continuous, and low-latency connectivity is therefore essential to support these applications around the clock.

Fortunately, wireless service providers are responding with new deployments, including the transition toward 5G networks. As a result, concerns about connectivity and data transfer speeds are gradually diminishing.

Instead, focus is shifting toward higher-level network properties such as privacy and security, especially as reliance on IoT expands into smart homes, smart cities, Industry 4.0, V2X, and many other aspects of daily life.

This new digital world enables companies and consumers to remain constantly connected. Highly connected IoT users spend much of their day online for work, study, entertainment, social media, and many other activities. Not only do users spend many hours online each day, but IoT devices themselves also operate continuously.

Sensors, cameras, and monitoring systems remain connected to the Internet at all times, continuously collecting information and helping users make better decisions about their environments and activities.

IoT generates enormous amounts of data because it operates continuously using sensors embedded in machines, environmental monitoring systems, and smart home devices. This makes IoT a major driver of large-scale data analytics, enabling

organizations to analyze information and making smart systems increasingly intelligent.

This data takes many forms, including voice commands, sensor readings, video streams, and more. IoT metadata is stored in databases to structure unstructured content or processed through cognitive systems to introduce new levels of intelligence, understanding, and decision-making into our environments.

The level of intelligence enabled by IoT can be remarkable. Many people are surprised by the number of IoT devices connected to their home networks. A modern smart home may easily contain 10 to 20 IoT devices, including cameras, smart appliances, lighting systems, smart speakers such as Amazon Alexa, smartwatches, and fitness trackers.

These devices operate continuously, monitoring environments, collecting data, communicating with each other, and supporting decision-making processes. A simple network scan can reveal just how deeply IoT has become integrated into everyday life.[1]

Section Three: Intrusion Detection Systems (IDS) :

Introduction to Intrusion Detection Systems (IDS)

The continuous evolution of malware represents a major challenge in the design of Intrusion Detection Systems (IDS). Cyberattacks have become increasingly sophisticated, and one of the greatest challenges is detecting unknown and obfuscated malware. Malware developers use advanced obfuscation techniques to hide malicious information and avoid detection by IDS solutions.

In addition, security threats such as zero-day attacks have increased significantly, directly targeting Internet users. As a result, cybersecurity has become an essential requirement in modern life, which now heavily depends on information technology.

Zero-day attacks have greatly affected countries such as Australia and the United States. According to the Symantec Internet Security Report 2017, more than three billion attacks of this type were reported in 2016, demonstrating that both the scale and severity of these attacks exceeded previous expectations. Furthermore, data breach statistics from 2017 revealed that nearly nine billion data records had been lost or stolen by cybercriminals since 2013.

Symantec reports also indicate that cybersecurity incidents continue to rise. In the past, cybercriminals mainly targeted bank customers to steal money from accounts or credit cards. However, modern malware has become more ambitious, directly targeting financial institutions and sometimes attempting to steal millions of dollars in a single attack. For this reason, detecting zero-day attacks has become a top priority.

Major cybercrime incidents have demonstrated how easily cyber threats can spread globally, where even a small security breach can disrupt vital services or infrastructure. Cybercriminals around the world are motivated by objectives such as information theft, illegal financial gain, and discovering new targets. Malware is intentionally designed to penetrate computer systems and exploit weaknesses in intrusion detection mechanisms. In 2017, the Australian Cyber Security Centre (ACSC) conducted a critical study on the different levels of sophistication used by attackers.

Therefore, there is an urgent need to develop effective intrusion detection systems capable of identifying emerging and complex malware. IDS aims to detect different forms of malicious software as early as possible, which traditional firewalls alone

cannot achieve. With the increasing amount of malware targeting computer systems, developing enhanced IDS solutions has become critically important.

Over the past few decades, machine learning techniques have been widely applied to improve intrusion detection. However, there remains a need for comprehensive and up-to-date classifications and surveys of recent work in this field. Many existing studies rely on datasets such as KDD-Cup 99 or DARPA 1999 to evaluate IDS development. Nevertheless, there is still no clear answer regarding which data mining techniques are most effective. Furthermore, the time required to build an IDS is often ignored, despite being a crucial factor for real-time (online) IDS effectiveness.

This paper presents an updated classification and review of major IDS research to date. It categorizes proposed systems according to modern IDS classifications and provides a structured overview of existing IDS approaches to help researchers quickly understand the main aspects of anomaly detection. The paper also surveys data mining techniques applied to IDS design.

Both Signature-based IDS (SIDS) and Anomaly-based IDS (AIDS) approaches are discussed, along with the techniques used in each method. The complexity of various AIDS methods and their evaluation techniques are analyzed, followed by recommendations for selecting the most appropriate approach depending on the nature of the intrusion.

The paper also discusses the challenges faced by modern IDS systems. Compared with previous survey studies, this work provides a broader discussion of IDS datasets, which represent a major concern for the Network Intrusion Detection Systems (NIDS) research community. Previous studies did not comprehensively review IDS systems in terms of datasets, challenges, and techniques simultaneously.

Several survey papers on intrusion detection have been published over the years. The survey by Axelsson (2000) classified IDS systems according to detection methods. Another widely cited survey by Debar et al. (2000) categorized detection techniques based on attack behaviors and characteristics. Later, Liao et al. (2013) introduced a classification consisting of five categories: statistical-based, pattern-based, rule-based, state-based, and heuristic-based approaches.

Unlike previous works, this research focuses on signature detection, anomaly detection, IDS classification methods, and datasets simultaneously.

Current review papers usually focus on a single aspect such as intrusion detection techniques, dataset problems, attack types, or IDS evasion techniques. No previous work has comprehensively reviewed intrusion detection, dataset challenges, evasion techniques, and attack categories together. Moreover, because IDS technologies continue evolving rapidly, many new systems have emerged, making updated surveys necessary.

This work therefore provides an updated survey and classification of intrusion detection systems while extending the classifications proposed in previous studies such as Liao et al. (2013) and Ahmed et al. (2016).

Based on the discussion of previous surveys, this work focuses on:

Classifying different IDS types according to intrusion methods.

Presenting evaluation metrics for anomaly-based network IDS and discussing the importance of feature selection.

Evaluating available IDS datasets and discussing challenges related to IDS evasion techniques.

An intrusion can be defined as any unauthorized activity that causes harm to an information system. Any attack threatening the confidentiality, integrity, or availability of information is considered an intrusion. For example, activities that make computer services unavailable to legitimate users are classified as intrusions.

An IDS is a software or hardware system that identifies malicious activities on computer systems in order to maintain system security. The goal of IDS is to identify malicious network traffic and unauthorized computer usage that traditional firewalls cannot detect. This capability is essential for protecting the availability, integrity, and confidentiality of computer systems.

IDS systems are generally classified into two major categories:

Signature-based Intrusion Detection Systems (SIDS)

Anomaly-based Intrusion Detection Systems (AIDS)

Signature-Based Intrusion Detection Systems (**SIDS**) :

Signature-based IDS relies on pattern matching techniques to identify previously known attacks. These systems are also referred to as knowledge-based detection or misuse detection systems.

In SIDS, matching methods are used to identify previously observed intrusions. In other words, when the signature of current activity matches a known attack signature stored in the signature database, an alarm is triggered.

SIDS systems examine host logs to identify sequences of commands or actions that have been predefined as malicious. In the literature, these systems are also known as knowledge-based or misuse detection systems.

The main concept of SIDS is to build a database of intrusion signatures and continuously compare current activities against stored signatures. If a match is found, the system generates an alert. For example, a rule-based signature may follow the format:

If (Source IP Address = Destination IP Address) → classify as an attack.

SIDS generally provides excellent accuracy when detecting previously known intrusions. However, it struggles to detect zero-day attacks because no corresponding signature exists in the database until the new attack signature is extracted and added.

SIDS techniques are used in many well-known tools such as Snort and NetSTAT.

Traditional SIDS approaches inspect network packets and attempt to match them against a signature database. However, these techniques are often unable to detect attacks distributed across multiple packets. Since modern malware has become increasingly sophisticated, extracting attack signatures across multiple packets may be necessary, requiring the IDS to analyze previous packet contents.

Signature creation techniques in SIDS may involve:

State machines, Formal language patterns, Semantic conditions

The increasing number of zero-day attacks has gradually reduced the effectiveness of traditional SIDS techniques because these attacks do not yet have predefined signatures. Furthermore, polymorphic malware and targeted attacks further weaken the efficiency of this traditional approach.

A possible solution to these limitations is the use of Anomaly-based Intrusion Detection Systems (AIDS), which detect attacks by identifying deviations from

normal behavior rather than relying on predefined signatures. This approach will be discussed in the following section.

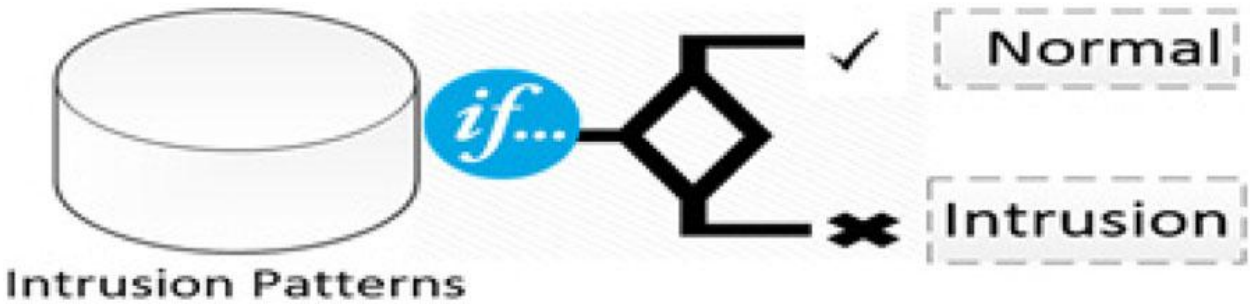

Figure 1-0-7 Conceptual Framework of SIDS Approaches

Anomaly-Based Intrusion Detection Systems (**AIDS**) :

Anomaly-Based Intrusion Detection Systems (AIDS) have attracted significant attention from researchers because of their ability to overcome many of the limitations of Signature-Based Intrusion Detection Systems (SIDS).

In AIDS, a model representing the "normal" behavior of a computer system is created using machine learning, statistical, or knowledge-based techniques. Any significant deviation between the observed behavior and the normal behavior model is considered an anomaly, which may indicate an intrusion.

These techniques are based on the assumption that malicious behavior differs from normal user behavior. Therefore, abnormal activities that do not conform to standard behavior patterns are classified as intrusions.

The development of AIDS typically consists of two stages:

- **Training Phase**:
  A profile of normal traffic behavior is used to learn and build the normal behavior model.

- **Testing Phase**:
  A new dataset is used to evaluate the system's ability to generalize and detect previously unseen intrusions.

AIDS systems can be classified into several categories depending on the training approach used, such as:

- Statistical-based systems
- Knowledge-based systems
- Machine-learning-based systems

Main Advantages of AIDS :

The most important advantage of AIDS is its ability to detect **zero-day attacks**, since anomaly detection does not rely on predefined signature databases. AIDS generates alerts whenever observed behavior deviates from normal behavior patterns.

In addition, AIDS provides several other important benefits:

- **Detection of Internal Malicious Activities**:
  If an attacker uses a stolen account to perform activities that differ from the legitimate user's normal behavior, the system can detect the anomaly and trigger an alert.
- **Resistance to Evasion**:
  It is very difficult for attackers to predict what the system considers "normal" behavior because the model is built using customized user profiles.

While SIDS can only detect well-known attacks, AIDS is capable of detecting previously unknown attacks and zero-day threats. However, AIDS may suffer from a

relatively high **False Positive Rate**, because some anomalies may represent new but legitimate activities rather than actual intrusions.

Due to the absence of a universal classification for anomaly-based IDS systems, they are commonly categorized into five subcategories according to their characteristics:

- Statistical-based
- Pattern-based
- Rule-based
- State-based
- Heuristic-based approaches

Techniques for Implementing Anomaly-Based IDS (AIDS) :

This section presents an overview of anomaly-based intrusion detection approaches proposed in recent years to improve detection accuracy and reduce false alarms.

AIDS techniques can generally be classified into three major groups:

1. **Statistics-Based Approaches**

These approaches collect and analyze records from datasets and build statistical models representing normal user behavior.

2. **Knowledge-Based Approaches**

These approaches identify expected activities using existing system knowledge, such as protocol specifications and network traffic states.

3. Machine–Learning–Based Approaches

These approaches learn complex pattern–matching capabilities from training data and are widely used in modern IDS systems.

Figure 2 illustrates these three main categories along with examples of their subcategories.

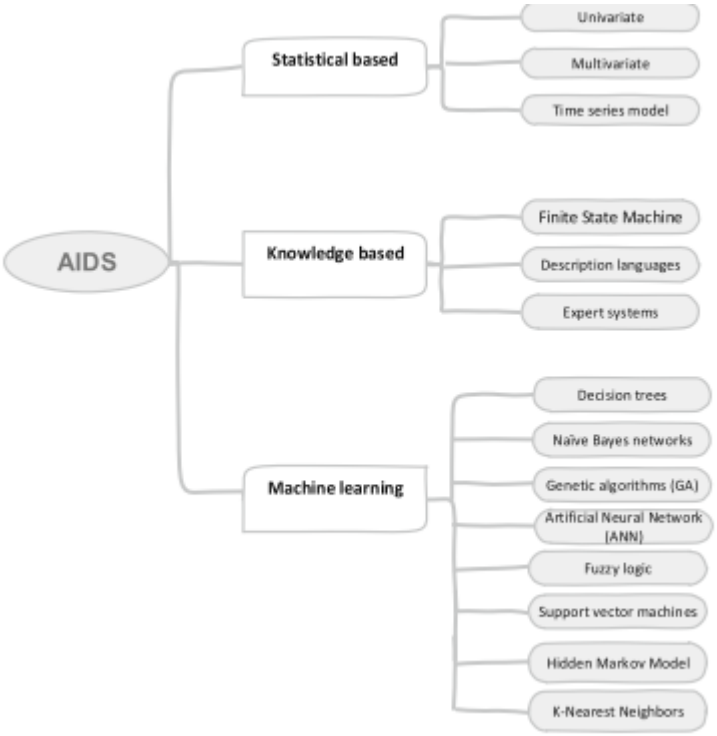

Figure 1-0-8 Classification of AIDS Approaches

Knowledge–Based Techniques :

This group of techniques is also known as the **Expert System approach**.

This method requires building a knowledge base that represents the profile of legitimate network traffic. Activities that differ from this standard profile are treated as intrusions.

Unlike other categories of anomaly-based IDS, the normal behavior model is usually created using human expertise in the form of rules that attempt to define normal system activities.

The main advantage of knowledge-based techniques is their ability to reduce **false-positive alarms**, since the system possesses knowledge of normal behaviors. However, in dynamically changing computing environments, this type of IDS requires regular updates to maintain information about expected normal behavior. This process is time-consuming because collecting information about all possible normal behaviors is extremely difficult.

Finite State Machine (**FSM**) :

A Finite State Machine (FSM) is a computational model used to represent and control execution flow. This model can be applied to intrusion detection to create an IDS model.

The model is typically represented through:

- States
- Transitions
- Activities

A state examines historical data. For example, changes in inputs are monitored, and based on the detected changes, transitions occur. FSM models can represent legitimate system behavior, and any noticeable deviation from this model is considered an attack.

Description Language:

Description languages define rule structures that can be used to specify characteristics of a particular attack.

Rules may be created using description languages such as:

- N-grammars
- UML

Expert Systems:

An expert system contains a collection of rules defining attacks.

In expert systems, these rules are usually manually defined by a knowledge engineer working alongside a domain expert.

Signature Analysis:

Signature analysis is one of the oldest techniques used in IDS systems. It relies on the simple concept of pattern matching.

In text matching, incoming packets are inspected word by word against known signatures. If a match is found, an alarm is triggered. Otherwise, traffic information is compared with the next signature stored in the signature database.

Supervised Learning in Intrusion Detection Systems (**IDS**):

This section presents different supervised learning techniques used in intrusion detection systems. Each technique is explained in detail with references to important research publications.

Supervised learning-based IDS techniques detect intrusions using labeled training data. This approach usually consists of two stages:

- Training
- Testing

During the training phase, relevant features and classes are identified, and the algorithm learns from these data samples.

In supervised IDS systems, each record is represented as a pair containing:

- Network or host data source
- Output label (classification), such as “intrusion” or “normal”

Feature selection may then be applied to remove unnecessary features.

Using the selected training features, supervised learning techniques train a classifier to learn the relationship between input data and labeled outputs.

A wide variety of supervised learning techniques have been explored in IDS research, each with its own strengths and weaknesses. During the testing phase, the trained model is used to classify unknown data as either “intrusion” or “normal.”

The resulting classifier becomes a predictive model capable of determining the most likely class for given feature values.

Figure 3 illustrates a general approach for applying classification techniques.

There are many classification methods used in IDS, including:

- Decision Trees
- Rule-Based Systems
- Neural Networks
- Support Vector Machines (SVM)

- Naïve Bayes
- Nearest Neighbor methods

Each technique uses a different learning strategy to build a classification model. However, a successful classification approach must not only handle training data effectively but must also accurately classify previously unseen records. Building classification models with strong generalization capability is therefore a key objective in machine learning.

Decision Trees :

A decision tree consists of three main components:

- **Decision Node**: Used to test a feature.
- **Branch**: Represents a possible decision outcome based on feature values.
- **Leaf**: Contains the class label of the item.

Several decision tree algorithms exist, including:

- ID3
- C4.5
- CART

Naïve Bayes :

Naïve Bayes applies Bayes' theorem while assuming strong independence between features.

It answers questions such as:

"What is the probability of a specific attack occurring given the observed system activities?"

using conditional probability formulas.

Naïve Bayes relies on features having different probabilities in attack and normal behaviors.

It is one of the most widely used classification models in IDS because of:

- Simplicity
- Computational efficiency

However, Naïve Bayes performs poorly when feature independence assumptions are violated, as observed in the KDD'99 intrusion detection dataset, which contains complex feature dependencies.

Studies have also shown that Naïve Bayes may have low accuracy on large datasets. More advanced variants such as **Hidden Naïve Bayes** (**HNB**) can better handle high-dimensional and highly correlated data in high-speed networks.

Genetic Algorithms (**GA**) :

Genetic Algorithms are heuristic optimization techniques inspired by biological evolution.

Each possible solution is represented as:

- A sequence of bits (genes)
- A chromosome

Solutions evolve over time through:

- Selection

- Reproduction
- Mutation

When applied to intrusion classification, chromosomes may represent:

- Clustering structures
- Integer-encoded cluster centers

Murray et al. used GA to develop simple network traffic rules. Each rule was represented as a genome containing genes corresponding to:

- Source IP
- Destination IP
- Source Port
- Destination Port
- Protocol Type

Artificial Neural Networks (**ANN**):

ANN is one of the most widely applied machine learning techniques in malware and intrusion detection.

The most common supervised learning technique used in ANN is the **Backpropagation** (**BP**) algorithm.

BP evaluates network error gradients relative to adjustable weights.

However, IDS systems based on ANN still face challenges in detecting less frequent attacks because training data for rare attacks is often limited compared to common attacks.

As a result:

- Detection accuracy becomes lower for rare attacks
- Low-frequency attacks may escape detection

This is particularly dangerous in cybersecurity because attacks such as **User–to–Root** (**U2R**) may grant attackers privileged access to victim systems.

ANNs may also suffer from:

- Local minima problems
- High training time

Nevertheless, ANNs are extremely powerful because hidden layers allow them to model highly nonlinear relationships between features and classifications.

Modern ANN variants include:

- Recurrent Neural Networks (RNN)
- Convolutional Neural Networks (CNN)

These models are widely used in IDS classification tasks.

Fuzzy Logic :

Fuzzy logic relies on degrees of uncertainty instead of traditional binary true/false logic.

It provides a direct way to reach conclusions based on:

- Imprecise
- Ambiguous
- Noisy
- Incomplete data

Using fuzzy domains allows elements to partially belong to multiple classes simultaneously.

Fuzzy logic is suitable for IDS because security problems inherently involve uncertainty, and the boundaries between normal and abnormal behaviors are not always clearly defined.

Traditional strict numerical boundaries often produce high false alarms. Slight deviations from normal behavior may either go undetected or generate unnecessary alerts.

Fuzzy logic can model these slight anomalies while maintaining lower false-positive rates.

Research by Elhag et al. demonstrated that fuzzy logic can reduce false alarm rates in intrusion detection by defining fuzzy rules describing normal and abnormal system activities.

Support Vector Machines (SVM) :

SVM is a discriminative classifier defined by a separating hyperplane.

SVM maps training data into higher-dimensional spaces using kernel functions so that intrusion classes become linearly separable.

SVMs are well known for:

- Strong generalization capability
- Effectiveness with high-dimensional data
- Good performance when the number of features is large

Different kernels can be used, including:

- Linear
- Polynomial
- Gaussian Radial Basis Function (RBF)
- Hyperbolic Tangent

In IDS datasets, many features may be redundant or have little effect on classification. Therefore, feature selection is important when training SVM models.

Li et al. applied an SVM classifier with an RBF kernel on the KDD 1999 dataset after carefully selecting a subset of features.

Hidden Markov Model (**HMM**) :

HMM is a statistical Markov model where the modeled system is assumed to be a Markov process with hidden states.

Research has shown that HMM can be applied to identify certain types of malwares.

In this technique:

- HMM is trained using malware features such as opcode sequences.
- Incoming traffic is then evaluated using the trained model.
- Results are compared with predefined thresholds.

If the score exceeds the threshold:

- The traffic is classified as malicious.

Otherwise:

- The traffic is considered normal.

K-Nearest Neighbors (**KNN**) :

KNN is a non-parametric classification method widely used in machine learning.

The idea is to classify an unlabeled sample according to the majority class among its k nearest neighbors.

For example:

- If k = 5
- Three neighbors belong to the "intrusion" class
- Two neighbors belong to the "normal" class

then the sample is classified as "intrusion" through majority voting.

KNN is commonly used as a benchmark classifier because it often provides good classification performance in IDS applications.

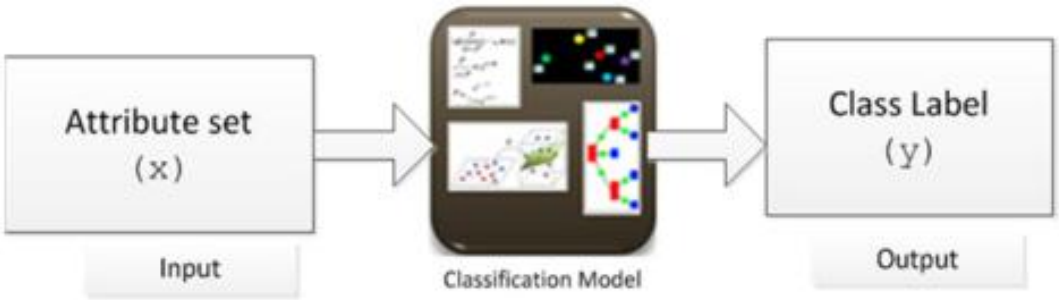

Figure 1-0-9 Classification as a Task

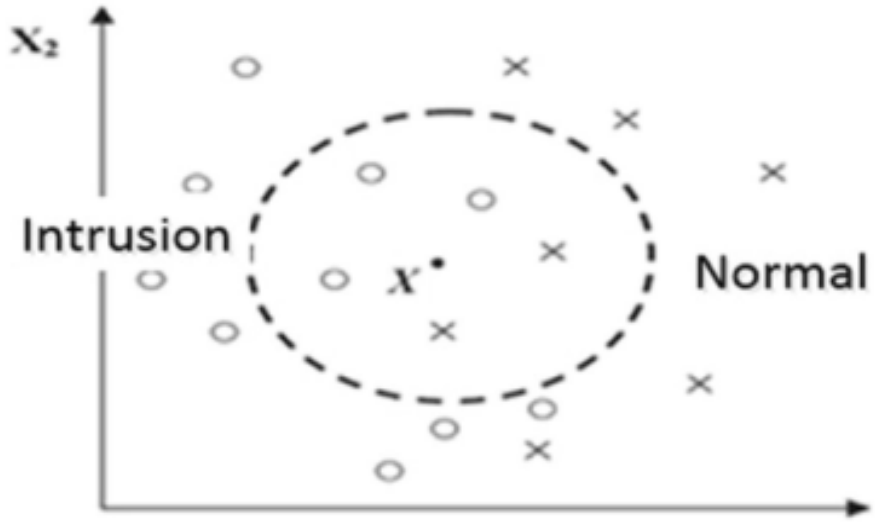

Figure 1-10 Example of K-Nearest Neighbor Classification for k = 5

Semi-Supervised Learning in Intrusion Detection Systems (IDS) :

Semi-supervised learning lies between supervised learning (which uses fully labeled training data) and unsupervised learning (which uses no labeled data).

Researchers have shown that semi-supervised learning can improve classifier performance in intrusion detection systems while reducing the time and cost required for labeling data. This advantage is particularly important because labeled intrusion detection datasets are often rare or difficult to obtain.

Several semi-supervised learning techniques have been proposed, including:

- Expectation Maximization (EM)
- Self-training

- Co-training
- Semi-Supervised Support Vector Machines (Semi-Supervised SVM)
- Graph-based methods
- Boosting-based semi-supervised learning approaches

Rana et al. proposed a new fuzzy-based semi-supervised learning approach that uses unlabeled samples combined with supervised learning algorithms to improve IDS classifier performance.

In this method:

- A feedforward neural network with a single hidden layer is trained to generate fuzzy membership vectors.
- Unlabeled samples are classified into fuzzy groups (low, medium, and high fuzziness).
- The classifier is retrained after integrating each fuzzy group separately into the original training set.

Experimental results using the NSL-KDD dataset showed that unlabeled samples belonging to low and high fuzziness groups significantly improved IDS accuracy compared with traditional methods.

Hybrid-Based Techniques :

Traditional IDS systems suffer from several limitations:

- Difficulty adapting to new environments
- Failure to detect new malicious attacks
- Low detection accuracy

- High false alarm rates

On the other hand, anomaly-based IDS (AIDS) alone may also suffer from high false-positive rates.

To overcome these limitations, Hybrid IDS systems were developed by combining both:

- Signature-Based IDS (SIDS)
- Anomaly-Based IDS (AIDS)

Hybrid systems combine the strengths of both approaches while reducing their weaknesses.

For example, Farid et al. proposed a hybrid IDS combining:

- Naïve Bayes
- Decision Trees

This model achieved a detection rate of 99.63% on the KDD'99 dataset.

Performance Metrics for Intrusion Detection Systems (IDS) :

Several classification metrics are used to evaluate IDS performance. IDS systems are commonly assessed using the following standard performance measures:

True Positive Rate (TPR) :

TPR is calculated as:

The ratio of correctly predicted attacks to the total number of attacks.

If all intrusions are detected, then:
[TPR = 1]

which is extremely rare for any IDS system.

TPR is also known as:

- Detection Rate (DR)
- Sensitivity

False Positive Rate (FPR) :

FPR is calculated as:

The ratio of normal instances is incorrectly classified as attacks to the total number of normal instances.

False Negative Rate (FNR)

A False Negative occurs when the IDS fails to identify an intrusion and incorrectly classifies it as normal traffic.

Classification Rate (CR) / Accuracy

Classification Rate measures how accurately the IDS detects both normal and abnormal traffic behavior.

It is defined as:

The percentage of all correctly predicted instances relative to the total number of instances.

Receiver Operating Characteristic (ROC) Curve

The ROC curve contains:

- X-axis → False Positive Rate (FPR)
- Y-axis → True Positive Rate (TPR)

In the ROC curve, TPR is plotted as a function of FPR for different decision thresholds.

Each point on the ROC curve represents:

- A specific FPR value
- A corresponding TPR value

As the classification threshold changes, different points on the ROC curve are selected with different False Alarm Rates (FAR) and TPR values.

An ideal classifier with perfect discrimination ability has an ROC curve passing through the upper-left corner, representing:

- 100% Sensitivity
- 100% Specificity.[1]

Section Four: Sensors Used in Smart Home Systems:

Sensors play a critical role in the development of Smart Home Systems (SHSs). During the literature review phase, several types of sensors were identified as commonly used in smart home applications.

This section provides a detailed overview of the sensors used in the design, development, and implementation of smart home systems.

1. Infrared Sensor (IR)

Theoretically, all objects and living organisms with a minimum temperature emit infrared radiation. An Infrared Sensor is an electronic device capable of detecting and measuring infrared radiation in its surrounding environment.

The primary purpose of infrared sensors is:

- Motion detection
- Temperature measurement

There are two categories of infrared sensors:

- Active IR Sensor
- Passive Infrared Sensor (PIR)

Among them, PIR sensors are the most widely used in smart home systems.

A PIR sensor detects changes in electromagnetic radiation levels in its surrounding environment without emitting radiation itself, unlike active IR sensors.

This sensor is mainly used for motion detection. When an object enters the sensor's range, the sensor measures differences in infrared radiation levels and identifies its presence.

In smart homes, PIR sensors are commonly used in:

- Intrusion alarm systems
- Human activity detection
- Elderly healthcare systems

However, PIR sensors are generally preferred for indoor use because environmental conditions such as rain or snow may affect their performance.

2. Temperature Sensor

Temperature sensors measure ambient temperature to ensure comfortable living conditions.

Several types of temperature sensors have been used in smart home systems. These sensors monitor room temperatures and allow other devices such as:

- Fans
- Air conditioners
- Heaters

to operate automatically based on sensor readings.

3. Humidity Sensor

Humidity sensors detect and measure changes in water vapor or humidity levels in the surrounding air.

One of the most common humidity sensors is the **DHT11**, which can also measure temperature.

In smart homes, humidity sensors are used for:

- Monitoring room humidity levels
- Home automation systems
- Early warning systems

4. Gas Sensor

Gas sensors detect the presence of specific gases within the surrounding air.

They are commonly used to detect dangerous gases such as:

- Liquefied Petroleum Gas (LPG)
- Propane
- Methane

- Carbon Monoxide
- Butane
- Alcohol vapors

These sensors are especially important in smart homes because they provide early warnings about hazardous gas leaks.

5. Smoke Sensor

A smoke sensor is an electronic device that detects smoke in its environment and activates an alarm when smoke is detected.

Smoke sensors are widely used as precautionary fire detection systems in smart homes. They are usually programmed to:

- Trigger alarms
- Send warning notifications via GSM messages or smart applications

6. Ultrasonic Sensor

Ultrasonic sensors emit and receive ultrasonic sound waves to measure distances to objects.

The process works by:

- Sending ultrasonic waves toward a target
- Receiving reflected waves
- Converting them into electronic signals

In smart home systems, ultrasonic sensors are used in:

- Automatic door systems

- Water level monitoring
- Basic home automation applications

7. **Flame Sensor**

A flame sensor is an electronic device specifically designed to detect the presence of fire within its operating range.

It usually responds by:

- Triggering alarms
- Activating warning systems

Smart home systems focused on safety use flame sensors for early fire detection and user alerts.

8. **Light Detection Sensor**

This sensor measures light intensity in a specific area.

It is a photoelectric device capable of converting ambient light energy into electrical energy.

Types of light sensors include:

- Photoresistors
- Photodiodes
- Phototransistors

In smart home systems, these sensors are commonly used for:

- Home lighting automation

In one specialized system, a light sensor was used to inform visually impaired individuals whether it was daytime or nighttime.

10. Accelerometer

An accelerometer is an electronic instrument used to accurately measure acceleration forces.

In smart home systems, accelerometers are used for:

- Detecting human movement
- Intrusion detection
- Monitoring patient movement and health conditions

11. Door Sensor

A door sensor detects whether a door is opened or closed and notifies the user accordingly.

It operates using a pair of electrical connectors that complete or disconnect a circuit when the door changes state.

Door sensors are widely used in smart homes for:

- Monitoring door status
- Elderly care systems
- General activity monitoring inside the home

12. Gyroscope

A gyroscope, or angular velocity sensor, is used to measure orientation and angular motion while maintaining stability.

In smart home systems, gyroscopes are used for:

- Detecting body posture
- Gesture control systems

Gyroscopes are often combined with accelerometers.

13. **Pulse Sensor**

A pulse sensor is capable of continuously detecting and monitoring human heart rate.

In smart homes, this sensor is mainly used in:

- Elderly healthcare systems

14. **Fluid / Water Detection Sensor**

Also known as a **Raindrop Sensor**, this electronic device detects the presence of water using an exposed conductive surface.

It is commonly used for:

- Water leakage detection
- Water level monitoring in tanks

15. **Camera**

Cameras and camera networks are increasingly used in smart home systems.

Researchers integrate cameras with:

- Image processing
- Machine learning
- Gesture recognition

- Artificial intelligence

to improve security and monitoring capabilities.

Cameras are also used for:

- Object detection
- Gesture recognition
- Live video monitoring

Although cameras provide reliable home security and monitoring, their main disadvantages are:

- High cost
- Maintenance complexity

16. Force–Sensing Resistor (FSR)

An FSR is an electronic device capable of measuring force, pressure, or mechanical stress.

FSRs use special materials whose electrical resistance changes when force is applied.

These sensors are used in smart homes for:

- Detecting resident activities
- Elderly healthcare systems

17. Flex Sensor

A flex sensor measures bending or flexing in objects.

It operates by changing electrical resistance when bent.

Flex sensors are commonly used in:

- Gesture-controlled systems

Other Types of Sensors:

Additional integrated sensor systems combine multiple sensors into a single device, such as:

- Bed Sensors
- Chair Sensors
- Posture Sensors

These systems often integrate:

- Motion sensors
- Accelerometers
- Gyroscopes .[2]

Section Five: Security of IoT Devices (Inherent Vulnerabilities of IoT Devices):

Recent studies indicate that despite the growing popularity of IoT devices, they generally suffer from weak security.

For a device to be considered secure in information and network security, it should provide the following properties:

- **Identification**:
  The ability to uniquely identify devices or users.
- **Confidentiality**:
  Ensuring information is accessible only to authorized users.

- **Integrity**:
  Maintaining data accuracy and trustworthiness during storage, transmission, and processing.
- **Non-Repudiation**:
  Preventing senders from denying their transmitted messages or authorized transactions.

However, IoT device design limitations create inherent vulnerabilities. One major limitation is the restricted computational resources of IoT devices, limiting their ability to implement strong security mechanisms.

This often results in:

- Weak data encryption
- Insecure network services
- Default passwords vulnerable to brute-force attacks

Additional vulnerabilities include:

- Lack of automatic security updates
- Dependence on manual patch installation
- Discontinued manufacturer support for older devices
- Massive collection of sensitive user data
- Integration into larger networks that may expose entire infrastructures to attacks

Compromising a single IoT device may provide attackers with access to valuable information or critical network components.

Categories of Cyberattacks:

Cyberattacks targeting IoT devices can generally be divided into four categories:

- Physical Attacks
- Network Attacks
- Software Attacks
- Data Attacks

Physical Attacks:

Physical attacks require direct physical access to devices or networks.

Examples include:

- Hardware modification
- Injecting malicious code

One example is the **Voice Squatting** attack discovered in 2018, targeting devices such as Amazon Echo and Google Home using voice commands that imitate legitimate commands.

Network Attacks:

Network attacks exploit vulnerabilities in communication systems without requiring physical access.

Examples include:

- **Traffic Analysis**:
  Extracting sensitive information from weakly protected traffic.

- **Malicious Node Attacks**:
  Selectively modifying or dropping packets.
- **Man–in–the–Middle** (**MITM**):
  Eavesdropping on communication between devices.
- **Denial of Service** (**DoS**):
  Overloading systems with excessive requests.

A famous example is the **Mirai Botnet attack** against Dyn DNS services in 2016, which compromised insecure IoT devices such as DVRs and cameras.

Software Attacks:

Software attacks involve malware used to:

- Manipulate data
- Steal information
- Launch DoS/DDoS attacks

Examples include:

- Mirai Botnet
- The 2015 Jeep vehicle hacking incident

Data Attacks:

IoT systems often rely on cloud computing, creating additional risks such as:

- Data inconsistency
- Unauthorized access
- Data breaches

Many of these attacks involve insider access to sensitive systems.

Denial of Service (DoS):

A DoS attack overwhelms a target device or system with invalid traffic or excessive requests, causing severe slowdown or complete shutdown.

In IoT environments, such attacks may render devices unusable and require reinstallation or replacement.

A Distributed Denial of Service (DDoS) attack uses multiple compromised devices simultaneously, as demonstrated in the Mirai Botnet attack.

Weak default passwords in IoT devices are one of the primary reasons these attacks succeed.

Man–in–the–Middle (MITM) :

MITM attacks occur when attackers place malicious nodes between two legitimate devices to intercept and manipulate communications.

For example:

- Device A sends data to Device B
- The attacker intercepts and modifies the communication before forwarding it

Successful MITM attacks may allow attackers to:

- Manipulate traffic flows
- Reconfigure networks
- Create fake identities
- Inject malicious information

Standardization and Regulation of IoT Devices:

The lack of standardized security guidelines during IoT development is one of the main reasons IoT devices remain insecure.

The European Union is currently discussing the development of the **European Cybersecurity Certification Framework** (**ECSCF**) to address these issues by creating certification ecosystems for IoT devices.

However, challenges remain. Although organizations provide security standards and certifications, many maintain close relationships with governments, manufacturers, and service providers, potentially affecting the neutrality and credibility of these standards and certifications.[3]

Chapter Two: Literature Review :

Amouri, Alaparthy, and Morgera [4] developed an IDS for IoT networks using supervised machine learning to profile normal node behavior and detect anomalies in network traffic. Doshi et al. [5] also applied machine learning algorithms to detect Distributed Denial of Service (DDoS) attacks in IoT environments by focusing on IoT-specific network behaviors such as packet timing and limited endpoints. Meidan et al. [6] employed deep learning techniques for detecting botnets in IoT ecosystems through behavioral analysis of connected devices. Brun et al. [7] proposed a deep learning-based approach using dense Random Neural Networks for network attack detection, particularly DoS attacks. In addition, Subba et al. [8] implemented a model based on feed-forward and backpropagation algorithms to detect and classify cyber-attacks such as DoS, Probe, User to Root, and Remote to User attacks.[9] Anthi et al. proposed a smart home Intrusion Detection System based on supervised learning, capable of classifying IoT devices and detecting attacks such as DoS,

MITM, and Replay attacks within a real smart home environment. .[10] In addition, Alsaedi et al. . introduced the TON_IoT dataset, which contains telemetry data, network traffic, and operating system logs representing both malicious and normal behaviors, and has been widely used for training and evaluating Machine Learning-based IDS models in IoT and IIoT environments. . [11] Furthermore, recent Deep Learning-based studies have focused on applying Deep Learning techniques for anomaly and cyber-attack detection in IoT networks using modern datasets such as CICIoT2023 to improve detection accuracy and reduce false alarms in smart home IDS systems.

Table 1 Literature Review

Title	year	Benefit for our Project
Cross layer-based intrusion detection based on network behavior for IoT	2018	We benefited from this study in the idea of using supervised machine learning to analyze the normal behavior of IoT devices and detect anomalous activities within the network, which is similar to the concept of the second model in our project for malicious traffic detection.
Machine learning DDOS detection for consumer internet of things devices	2018	We benefited from this study in understanding the characteristics of DDoS/DoS attacks in IoT

		networks, especially features related to packet rate and packet timing, which helped us select suitable features for DoS attack classification.
N-baiotnetworkbased detection of IoT botnet attacks using deep autoencoders	2018	We benefited from this study in the concept of analyzing the behavioral patterns of IoT devices and creating behavioral profiles to detect abnormal activities and botnet attacks within IoT environments.
Deep learning with dense random neural network for detecting attacks against IoT-connected home environments	2018	We benefited from this study in understanding how Deep Learning techniques can be applied for network attack detection, especially DoS attacks, as well as the importance of extracting relevant network features before the training process.
A neural network-based system for intrusion detection and attack classification	2016	We benefited from this study in the idea of classifying different attack types rather than only

		detecting malicious traffic, which is similar to the functionality of the third model in our proposed system for attack classification.
TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data Driven Intrusion Detection Systems	2020	This study is considered one of the closest works to our project, as it proposed a smart home IDS based on supervised learning capable of detecting attacks such as: • DoS • MITM • Replay It also included IoT device classification within a real smart home environment, which is highly similar to the three-model architecture used in our project.
A comprehensive survey on techniques, challenges, evaluation metrics and applications of deep learning models for anomaly detection	2025	We benefited from this study through the use of the TON_IoT dataset introduced by the authors, which we used to train the IoT device

		identification model and analyze normal behavior within the smart home network.
A Supervised Intrusion Detection System for Smart Home IoT Devices	2019	We benefited from these studies in understanding the use of modern datasets such as: • CICIoT2023 for training IDS models capable of detecting modern cyber-attacks in IoT environments while improving: • Accuracy • Detection Rate • Reducing False Positives These studies also supported the development of the attack classification model in our proposed system.

Chapter Three: Smart Home Intrusion Detection System Using Machine Learning for Behavioral Analysis

3-1 Proposed System Overview

The proposed system aims to develop a smart Intrusion Detection System dedicated to smart home IoT environments. The system monitors network traffic and analyzes the behavior of IoT devices in order to detect malicious activities and cyber-attacks.

The system relies on Machine Learning algorithms to analyze network data and classify it as either normal behavior or malicious activity, with the ability to identify the type of attack occurring inside the network.

3-2 Working Environment

A virtual smart home environment was designed consisting of several IoT devices connected through a Wi-Fi network using an Access Point and a Gateway.

The environment includes multiple smart devices operating with different protocols such as MQTT and HTTP. Node-RED was used to simulate the operation of these devices and their communication within the network.

3-3 Proposed System Architecture

The proposed system consists of three main stages:

1- Data Collection Stage

This stage involves collecting network traffic data and IoT device behavior from the smart home environment, in addition to using specialized IoT security datasets.

2- Data Processing Stage

This stage includes data cleaning, removing unnecessary values, and extracting important features used during the training and detection processes.

3– Detection and Classification Stage

In this stage, Machine Learning algorithms are used to analyze the data, detect malicious activities, and classify attack types.

3-4 Used Datasets

The project relies on two main datasets:

Table 2 Used Datasets

Dataset	Purpose
TON_IoT	Device behavior analysis
CICIoT2023	Attack analysis and classification

These datasets were selected because they contain recent IoT-related network traffic and various cyber-attack scenarios.

3–5 Used Algorithms

Several Machine Learning algorithms were used in the proposed system, including:

Table 3 Used Algorithms

Algorithm	Purpose
Random Forest	Malicious activity detection
XGBoost	Attack type classification

These algorithms were selected because of their high performance in network traffic analysis and classification tasks.

3–6 Targeted Attacks

The proposed system focuses on detecting several common attacks in IoT networks, including:

- Denial of Service (DoS)
- Spoofing Attacks
- Replay Attacks
- Man in the Middle (MITM) Attacks

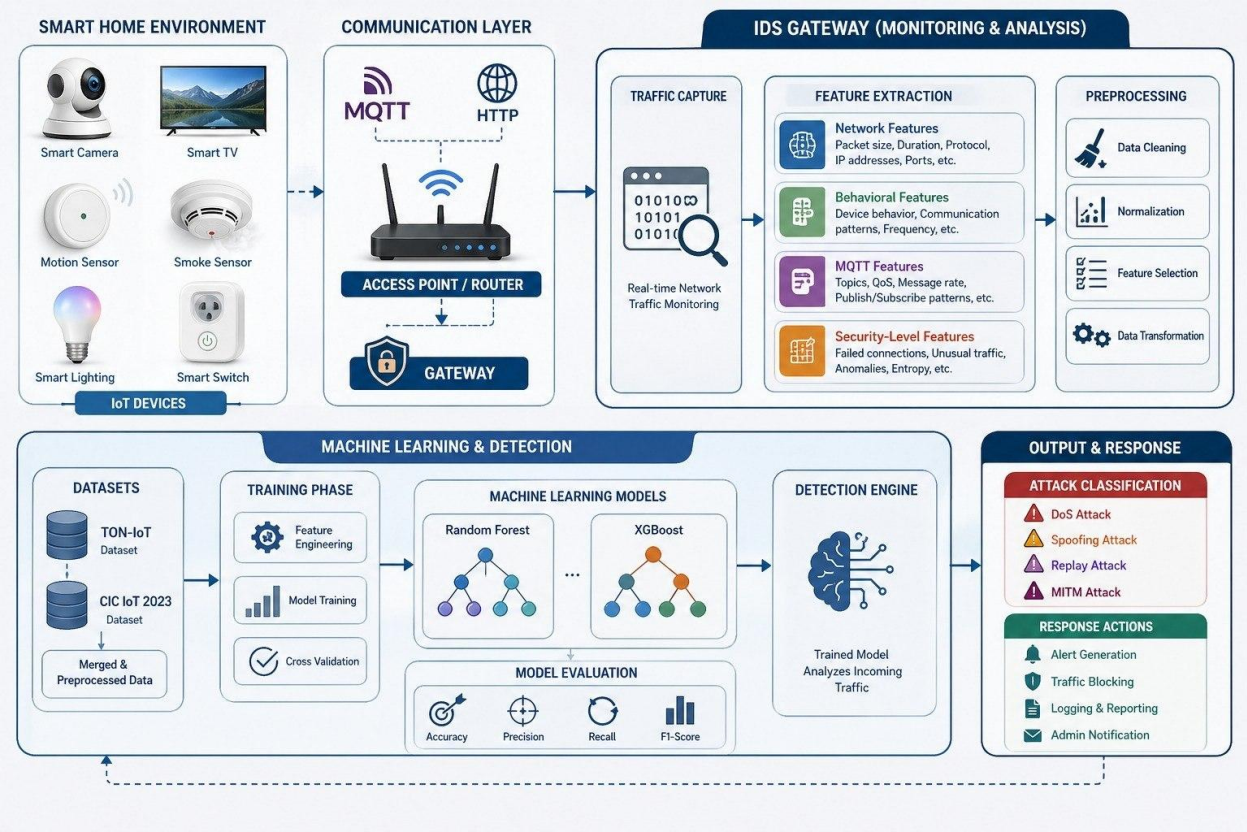

Figure 3 – 1 Proposed Smart Home IoT IDS Architecture

Chapter Four: Practical Implementation

4.1 Work Environment

The project was implemented inside a virtual environment using VMware Workstation Pro, where an Ubuntu operating system was created to serve as the main environment for simulating the smart home network and implementing the Intrusion Detection System (IDS).

Table 4-1 Work Environment Specifications

Component	**Specifications**
Operating System	Ubuntu Linux
Virtualization Software	VMware Workstation Pro
Programming Language	Python
Simulation Tools	Node-RED
Communication Protocols	MQTT – HTTP
Machine Learning Tools	Scikit-learn – Pandas – NumPy

4.2 Smart Home Network Design

A smart home network was designed containing several IoT devices connected through an Access Point / Router. The network traffic was monitored by an IDS gateway responsible for traffic analysis and attack detection.

The environment included the following devices:

- Smart Camera
- Smart TV
- Motion Sensor
- Smoke Sensor
- Smart Lighting
- Smart Switch

MQTT and HTTP protocols were used to simulate different communication patterns between the devices.

4.3 Simulation Environment Setup Using Node-RED

Node-RED was used to simulate the behavior of smart home devices and generate data flows representing IoT activities.

Several flows were created to simulate:

- Sensor data transmission.
- Smart device ON/OFF operations.
- MQTT message publishing.
- Normal network communication behavior.
- Normal and abnormal traffic scenarios.

This environment helped generate realistic network traffic that could later be used for training the intrusion detection models.

4.4 Datasets Used

Two main datasets were used in this project:

1. TON-IoT Dataset

This dataset was used for training the IoT device identification model and analyzing normal device behavior within the smart home network.

The dataset contains:

- Sensor data.
- IoT device data.

- Telemetry data.
- Network traffic data.
- Normal and malicious activities.

2. **CICIoT2023 Dataset**

This dataset was used for training the cyberattack detection and attack classification models.

It includes attacks such as:

- DoS
- MITM
- Spoofing
- Recon
- Replay-like behaviors

4.5 Data Preprocessing

Before training the models, several preprocessing steps were applied to improve data quality and prepare the datasets for machine learning.

These steps included:

- Data Cleaning.
- Removing missing values.
- Removing invalid values.
- Converting categorical values into numerical values.
- Feature Encoding.

- Data Normalization.
- Feature Selection.

The datasets were then divided into:

- 80% Training Data
- 20% Testing Data

4.6 Models Used in the System

The proposed IDS system was built using three main models:

table 4-2 1Proposed Machine Learning Models and Their Function

Model	**Function**
Model 1	IoT Device Identification
Model 2	Malicious Traffic Detection
Model 3	Attack Type Classification

4.7 Model 1: IoT Device Identification Model

The first model aims to identify the type of IoT device connected to the network based on behavioral characteristics and the generated device data. The idea behind this model is that each device inside the smart home has a unique communication pattern in terms of transmission rate, data type, and network behavior. Therefore, these characteristics can be used to create a unique "Device Fingerprint" for each IoT device.

The model was trained using data extracted from the TON-IoT dataset, which included data from different devices such as:

- Thermostat
- Fridge
- Motion Light

Several features were used, including:

- current_temperature
- thermostat_status
- motion_status
- fridge_temperature
- light_status
- temp_condition

After preprocessing and converting categorical values into numerical values, the dataset was divided into:

- 80% training data.
- 20% testing data.

The total number of records was:

- 1,116,684 data records.

The number of main features used for training was:

- 14 features.

Model 1 Results

The training results showed excellent model performance, as the model successfully distinguished between different IoT devices with very high accuracy.

Classification Report Results

table 4-2 2 Classification Results of the IoT Device Identification Mode

Device	Precision	Recall	F1-Score
Fridge	1.00	1.00	1.00
Motion Light	1.00	1.00	1.00
Thermostat	1.00	1.00	1.00

Overall Accuracy

- Accuracy = 100%

Results Analysis

These results indicate that the model was able to almost perfectly distinguish between different devices without any classification errors in the testing dataset.

The Confusion Matrix also showed that the model did not confuse device categories, indicating that the behavioral characteristics of each device were sufficiently unique to create an accurate behavioral fingerprint.

The importance of this model lies in forming the first layer of the intrusion detection system, as it allows the system to verify the identity of connected devices. Therefore, if a device behaves differently from its normal fingerprint, this may indicate a Spoofing Attack or abnormal activity.

```
Dataset shape: (1116684, 18)
Columns: ['current_temperature', 'thermostat_status', 'motion_status', 'fridge_temperature', 'device_Fridge', 'device_Motion_Light', 'device_Thermostat', 'light_status_0', 'light_status_ off', 'light_statu
s_ on', 'temp_condition_0', 'temp_condition_high', 'temp_condition_high ', 'temp_condition_high  ', 'temp_condition_low', 'temp_condition_low ', 'temp_condition_low  ', 'label']

Device distribution:
Fridge         486213
Thermostat     407176
Motion_Light   223295
Name: count, dtype: int64

X shape: (1116684, 14)
Train size: (893347, 14)
Test size: (223337, 14)

Training Model 1 (Device Identification)...

== MODEL 1 RESULTS ==

Confusion Matrix:
[[97243     0     0]
 [    0 44659     0]
 [    0     0 81435]]

Classification Report:
              precision    recall  f1-score   support

       Fridge       1.00      1.00      1.00     97243
Motion_Light       1.00      1.00      1.00     44659
  Thermostat       1.00      1.00      1.00     81435

    accuracy                           1.00    223337
   macro avg       1.00      1.00      1.00    223337
weighted avg       1.00      1.00      1.00    223337
```

figure 4 – 1 Results of the IoT Device Identification Model

4.8 Model 2: Malicious Traffic Detection Model

The second model aims to distinguish between benign traffic and malicious traffic within the smart home network.

This model relies on analyzing network and behavioral features of communications in order to determine whether the activity represents a cyberattack or normal behavior.

The model was trained using data containing:

- Normal traffic data.

- Attack traffic data.
- Multiple network-related features.
- Behavioral features related to IoT devices.

The dataset size was:

- 300,000 data records.

After encoding and preprocessing, the dataset contained:

- 82 features.

The dataset was divided into:

- 240,000 training samples.
- 60,000 testing samples.

Model 2 Results

Confusion Matrix

The results showed that the model successfully detected most attacks correctly with only a small number of classification errors.

Classification Report Results

table 4-2 3 Classification Results of the Malicious Traffic Detection Mode

Class	Precision	Recall	F1-Score
Normal (0)	0.83	0.90	0.86
Attack (1)	0.99	0.97	0.98

Overall Results

table 4-2 4 Performance Metrics of the Malicious Traffic Detection Mode

Metric	Value
Accuracy	97%
Macro Avg F1-Score	92%
Weighted Avg F1-Score	97%

Results Analysis

The results demonstrate that the model has a very strong capability to detect malicious activities within the network.

The model achieved a high Recall value of 97% for attacks, meaning that most attacks were successfully detected without being ignored.

In addition, the model achieved a very high Precision value of 99% for malicious traffic, indicating a low False Positive rate.

As for normal traffic, the results were slightly lower compared to malicious traffic due to the similarity between some normal and malicious behaviors in IoT environments.

These results confirm the effectiveness of the model as a primary detection layer in the proposed IDS system.

```
Dataset shape: (300000, 57)
X shape after encoding: (300000, 82)
Any NaN left? False
Any +inf left? False
Train size: (240000, 82)
Test size: (60000, 82)

Training model...

Confusion Matrix:
[[ 6376   686]
 [ 1332 51606]]

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.90      0.86      7062
           1       0.99      0.97      0.98     52938

    accuracy                           0.97     60000
   macro avg       0.91      0.94      0.92     60000
weighted avg       0.97      0.97      0.97     60000

Press any key to continue . . .
```

Figure 4 – 2 Results of the Malicious Traffic Detection Model

4.9 Model 3: Attack Classification Model

After detecting malicious activity, the third model is responsible for identifying the specific type of cyberattack occurring within the network.

This model is considered one of the most important components of the system because it not only detects attacks but also identifies the attack type in order to support appropriate security responses.

The model was trained using data containing the following classes:

- DoS
- MITM
- Spoofing
- Recon
- Benign
- Other

The dataset size was:

- 300,000 records.

The dataset contained:

- 40 main features.

Model 3 Results

Classification Report Results

table 4-2 5 Classification Results of the Attack Classification Mode

Attack Type	Precision	Recall	F1-Score
Benign	0.75	0.90	0.82
DoS	1.00	1.00	1.00
MITM	0.79	0.70	0.74
Other	0.94	0.93	0.94
Recon	0.75	0.44	0.55

Attack Type	Precision	Recall	F1-Score
Spoofing	0.75	0.58	0.65

Overall Results

table 4-2 6 Performance Metrics of the Attack Classification Mode

Metric	Value
Accuracy	97%
Macro Avg Recall	76%
Weighted Avg F1-Score	97%

Results Analysis

The results showed that the model achieved excellent performance in classifying DoS attacks, as it was able to detect them with almost perfect accuracy due to the clear characteristics of this type of attack, such as the significant increase in packet transmission rate and network requests.

The model also achieved good results in classifying MITM and Spoofing attacks. However, the performance was lower compared to DoS attacks because of the behavioral similarity between some attacks and normal network activities.

Recon attacks achieved the lowest Recall value due to several reasons:

- Limited training samples for Recon attacks.
- Similarity to normal behaviors.

- Class imbalance within the dataset.

Nevertheless, the model maintained a high overall accuracy of 97%.

These results indicate that the proposed system is capable of:

- Detecting cyberattacks.
- Identifying attack types.
- Supporting more accurate security decision-making.

```
Dataset shape: (300000, 40)

Attack distribution:
attack_type
DoS          238430
Other         37693
Benign        14057
Recon          4040
MITM           3617
Spoofing       2163
Name: count, dtype: int64
Train size: (240000, 39)
Test size: (60000, 39)

Training attack classification model...

Confusion Matrix:
[[ 2521     0    12   240    29     9]
 [    0 47683     0     2     0     1]
 [  122     0   509    18     8    66]
 [  372    34    11  7036    79     7]
 [  281     0     9   161   357     0]
 [   51     0   104    23     6   249]]

Classification Report:
              precision    recall  f1-score   support

      Benign       0.75      0.90      0.82      2811
         DoS       1.00      1.00      1.00     47686
        MITM       0.79      0.70      0.74       723
       Other       0.94      0.93      0.94      7539
       Recon       0.75      0.44      0.55       808
    Spoofing       0.75      0.58      0.65       433

    accuracy                           0.97     60000
   macro avg       0.83      0.76      0.78     60000
weighted avg       0.97      0.97      0.97     60000

Press any key to continue . . .
```

Figure 4 - 3 Results of the Attack Classification Model

4.10 Model Performance Evaluation

The three models were evaluated using several standard metrics commonly used in intrusion detection systems.

1. Accuracy

Accuracy measures the ratio of correctly predicted instances compared to the total number of instances.

The results showed that all models achieved very high accuracy rates:

- Model 1 → 100%
- Model 2 → 97%
- Model 3 → 97%

2. Precision

Precision measures how many detected attacks were actually true attacks.

A high Precision value indicates a low False Positive rate.

Model 2 achieved a very high Precision value for attacks:

- 99%

Meanwhile, Model 3 achieved excellent Precision for DoS attacks.

3. Recall

Recall measures the ability of the system to detect all actual attacks.

High Recall is extremely important in IDS systems because missing attacks represents a major security risk.

Model 2 achieved a high Recall value of:

- 97%

Model 3 also achieved perfect Recall for DoS attacks.

4. F1-Score

The F1-Score represents the balance between Precision and Recall.

The results showed that the models achieved a good balance between accurate detection and minimizing classification errors.

5. Confusion Matrix

The Confusion Matrix was used to analyze classification errors and identify which attack types caused the highest error rates.

The results showed that:

- Model 1: achieved nearly perfect classification.
- Model 2: produced very few classification errors.
- Model 3: experienced greater difficulty with low-sample attacks such as Recon and Spoofing.

The integration of multiple models within the proposed IDS system significantly improved analysis accuracy and reduced the probability of missing attacks or incorrectly classifying them.

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